SONUS MINIMAL

Single-ended Class A MOSFET amplifier — Build Lab Book

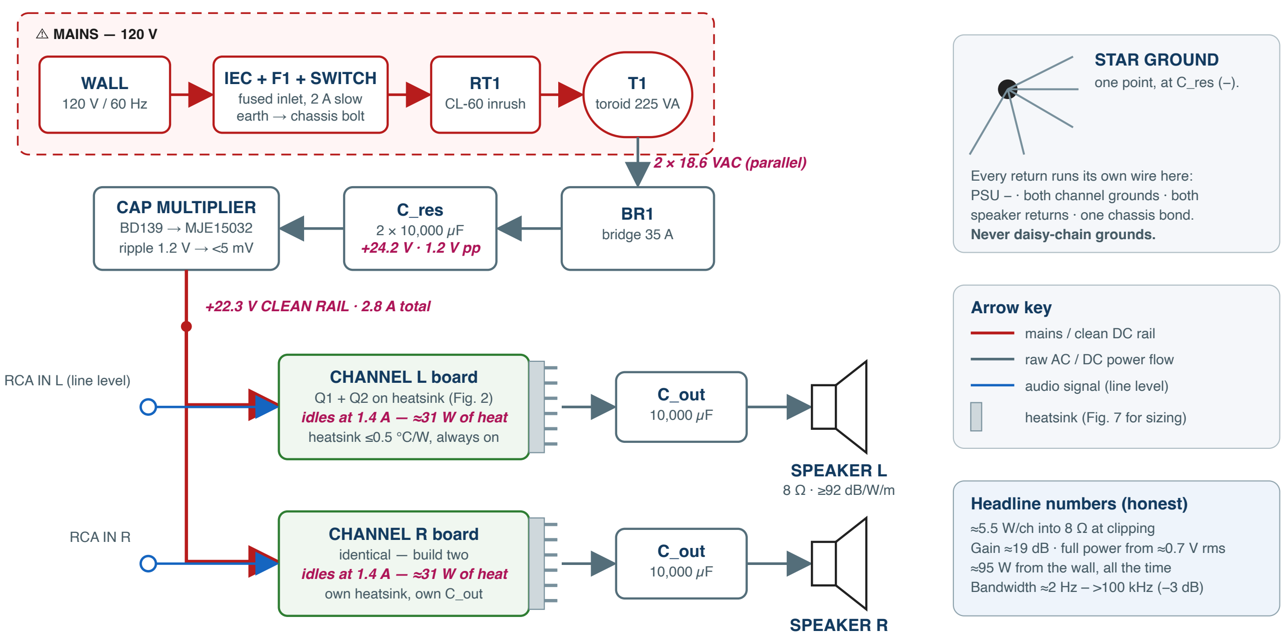
~5.5 W per channel into 8 Ω · Pass ACA lineage · three units (SM-001…003)

Document rev A · 2026-07-03 · design rev 4 (ECR-1…8 applied)

120 V / 60 Hz build · companion files: `diagrams/fig01–fig10`, `docs/BOM-rev4.csv`, `spice/*.cir`

Fig. 1 — System Block Diagram (one stereo amplifier)

One PSU feeds two identical channels · everything left of the transformer is lethal · everything right of it is warm but safe



SONUS MINIMAL — SE Class A 5 W	Fig. 1 — System block diagram (rev A) Supersedes 01_system_block.png	Sheet 1/1	2026-07-03 · §1

SONUS MINIMAL — Build Lab Book

Single-ended Class A MOSFET amplifier · ~5.5 W/channel into 8 Ω · Pass ACA lineage Document: LABBOOK rev A · 2026-07-03 · supersedes nothing (first edition) **Design baseline:** `docs/original/Sonus-Minimal-handoff.md` (BOM rev3) → resolved here as **rev 4 Mains:** 120 V / 60 Hz (US). 230 V differences are flagged where they occur.

Unit serial	Builder	Started	Completed	Sign-off (Phase G)
SM-001				
SM-002				
SM-003				

Contents

§0 How to use this book

Work **strictly in phase order**. Each phase ends with a gate — a short checklist that must be fully true before you continue. Skipping a gate is how amplifiers explode.

Conventions:

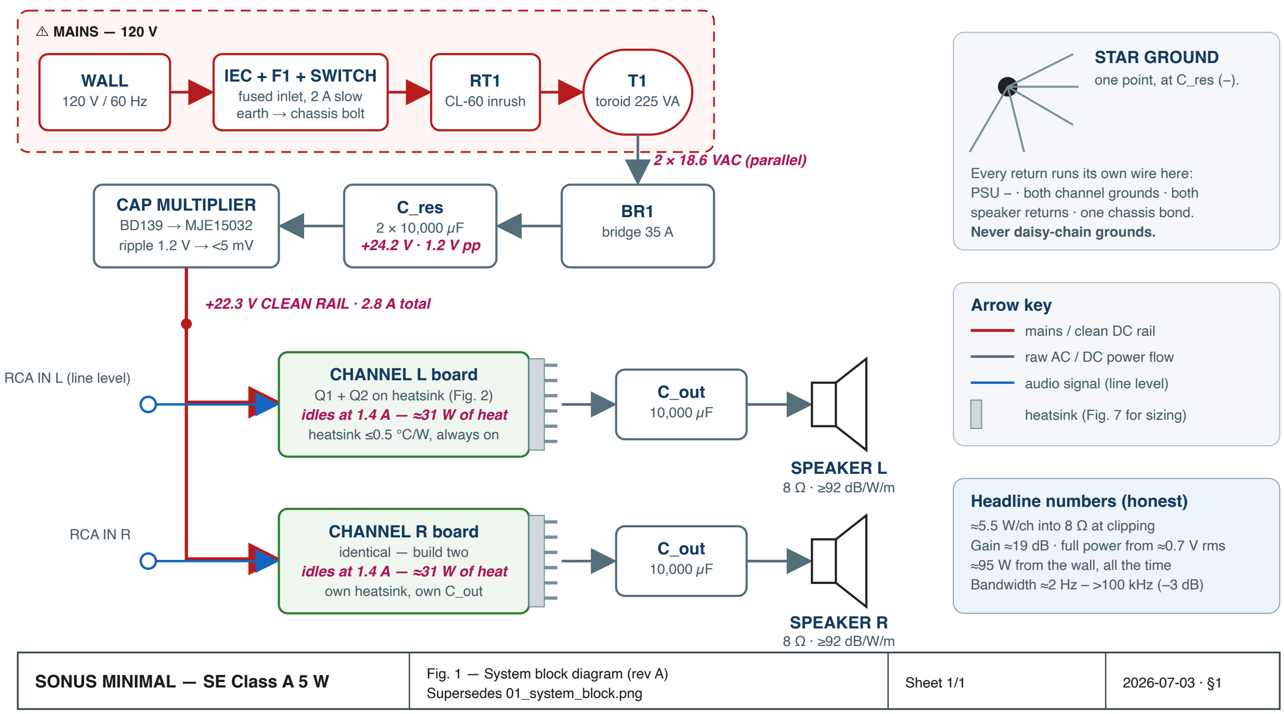
- `□` is a step you check off. **Record boxes** look like `V = ______` — write measured values in ink, in this book (or the per-unit logs in Appendix A). A lab book with empty boxes is a story; a lab book with numbers is evidence.
- ⚠ **DANGER** — mains or stored-energy hazard. Read twice, then act once.
- ⏱ — a mandated wait (thermal settle, capacitor discharge). Do not shortcut them.
- 📖 — a pointer into the reading list (§2) if you want the *why* behind the step.
- **Figures** live in `diagrams/` as SVG (`fig01`–`fig10`). Print Fig. 2, Fig. 3, and Fig. 5 and keep them at the bench.
- Every expected value in this book assumes the **rev 4** design (§4) and a 22.3 V rail. The original documents say "24 V" — §3/ECR-4 explains why the honest number is lower and why that is fine.

Time budget (per amplifier, experienced beginner): A: 3 h (once, covers all three amps) · B: 4 h · C: 5 h · D: 3 h · E: 2 h · F: 3 h · G: 2 h → ≈ **20 h for the first unit**, roughly half that for each following one.

§1 What you are building

Fig. 1 — System Block Diagram (one stereo amplifier)

One PSU feeds two identical channels · everything left of the transformer is lethal · everything right of it is warm but safe



A stereo, single-ended, **Class A** MOSFET amplifier in the Nelson Pass *Amp Camp Amp* / First Watt tradition. Two transistors per channel carry a constant 1.4 A; the speaker gets the *variation* in that current. There is no crossover distortion because there is no crossover — one device chain does all the work, all the time. The price is heat: the amp burns ~95 W from the wall whether it is whispering or working.

Design intent (from the original write-up, unchanged): *one dominant, smoothly decaying 2nd-harmonic coloration and the absence of everything that sounds hard — crossover artifacts, high-order products, hum, subsonic rumble.*

Honest specification (rev 4):

Parameter	Value	Notes
Output power, 8 Ω	≈ 5.5 W at clipping onset	current-limited to ≈3.9 W into 4 Ω
Class A envelope	up to ≈ 7 W (8 Ω)	soft 2nd-harmonic clipping beyond
Rail	**+22.3 V ± 1 V**	"24 V" in older docs — see ECR-4
Standing bias	1.40 A per channel	0.658 V across R_s1
Voltage gain	≈ 19–21 dB (measure yours)	full power from ≈ 0.65 V rms
Output impedance	≈ 0.6–0.8 Ω (DF ≈ 10–13)	source-follower + light feedback

Parameter	Value	Notes
Bandwidth	≈ 2 Hz – >100 kHz (–3 dB)	LF corner set by C_out into 8 Ω
Hum & noise	< 100 µV at output (target)	cap-multiplier rail, star ground
THD @ 1 W, 1 kHz	≈ 0.1–0.5 %, H2-dominant	this is the *design goal*, not a defect
Power from wall	≈ 95 W continuous	heat, always — see the two non-negotiables
Speakers required	**≥ 92 dB/W/m, ideally 95+**	non-negotiable #1
Heatsinking	**≤ 0.5–0.6 °C/W per channel**	non-negotiable #2

The two non-negotiables are inherited verbatim from the original design: this amplifier on an 85 dB bookshelf speaker is a disappointment by construction, and on an undersized heatsink it is a fire hazard by construction. Neither can be fixed downstream.

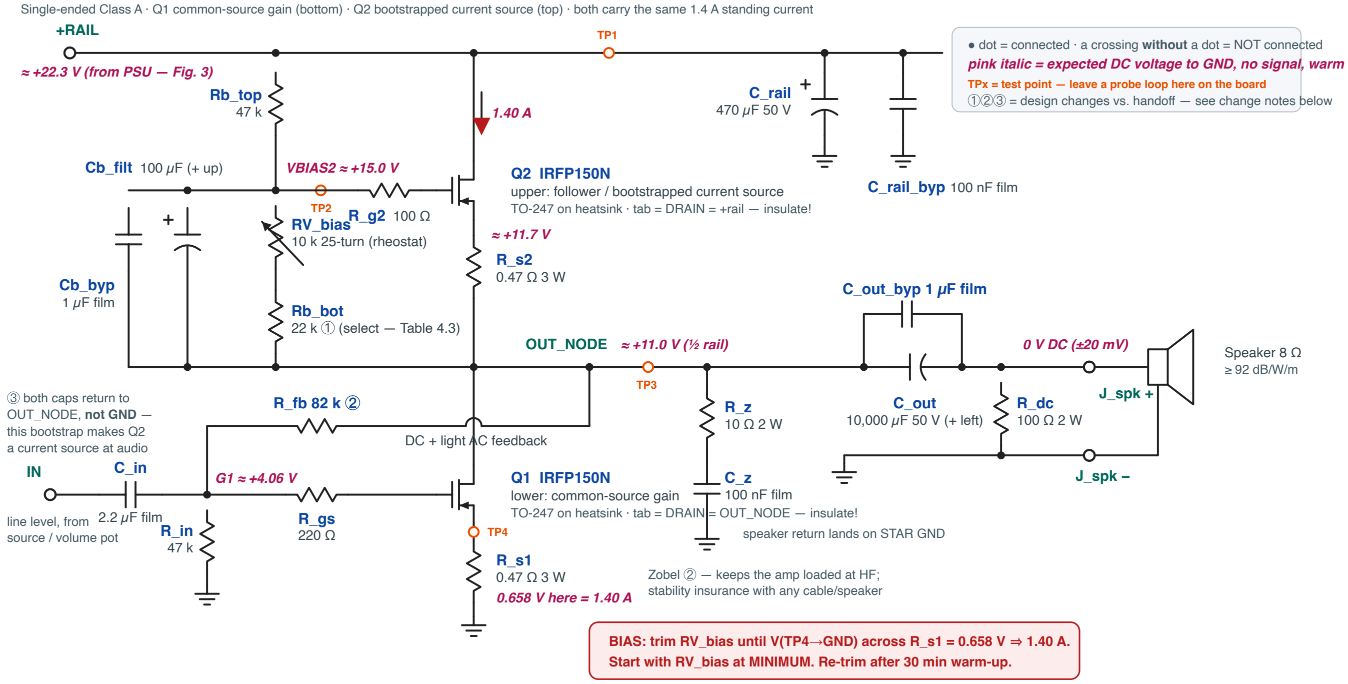
§2 Theory in thirty minutes + reading list

You can build this amp by following the steps alone. Read this section anyway — every debugging decision in Phases E–F gets easier when you know what each part is *for*.

2.1 The output stage: two identical MOSFETs, two different jobs

Fig. 2 — Channel Schematic (one of two identical channels) · rev A (resolved)

Single-ended Class A · Q1 common-source gain (bottom) · Q2 bootstrapped current source (top) · both carry the same 1.4 A standing current



① Rb_bot 10 k → 22 k (default; select 18 k / 22 k / 33 k from measured V_GS — Table 4.3). With the divider returned to OUT_NODE this centres the trim range for the IRFP150N. (ECR-2)
② R_fb 47 k → 82 k re-centres OUT_NODE at ½ rail for the IRFP150N's lower V_GS; Zobel R_z + C_z added for HF stability margin. (ECR-2, ECR-8)
③ Cb_filt / Cb_byp bootstrap to OUT_NODE is what turns Q2 into a quiet current source — wiring them to GND instead is the #1 way to build a "working" amp with no gain.
Expected DC map with a 22.3 V rail and matched FETs. Your OUT_NODE may sit anywhere in 10.3–12.3 V depending on V_GS — that is normal and correct (see §4.2).

SONUS MINIMAL — SE Class A 5 W	Fig. 2 — Channel schematic, resolved values (rev A) Supersedes 02_channel_schematic.png (IRFP240-era labels)	Scale: none · Sheet 1/1	2026-07-03 · LABBOOK §4

The rail-to-ground stack is: `+22.3 V → Q2 → R_s2 → OUT_NODE → Q1 → R_s1 → GND`.

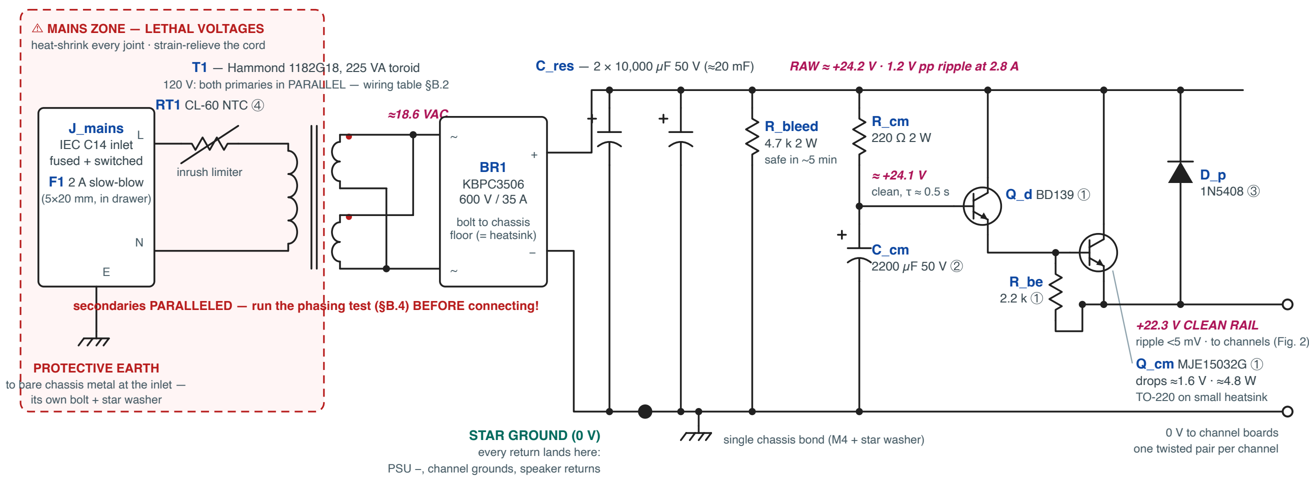
- **Q1 (bottom) is the voltage amplifier.** Common-source: the input signal wiggles its gate, and it responds by conducting more or less of the standing current. Its gain is approximately `gm·R_load / (1 + gm·R_s1)` ≈ 10–13× with an 8 Ω load. 📖 AoE §3.2.
- **Q2 (top) is a current source disguised as a follower.** Its gate is biased by the Rb_top / RV_bias / Rb_bot divider, and — this is the essential trick — that bias node is **bootstrapped to OUT_NODE through Cb_filt ∥ Cb_byp**, not bypassed to ground. At audio frequencies Q2's gate-to-source voltage therefore never changes, which makes Q2 a constant 1.4 A current source hovering over Q1. All of Q1's signal current has nowhere to go but the speaker. Wire those two capacitors to ground instead and you get a "working" amp with a gain of 0.3 — the single most common way to mis-build this circuit. 📖 Pass, *Amp Camp Amp*; AoE §2.2.5 (current sources), §2.4 (bootstrapping).

- **R_s1, R_s2 (0.47 Ω)** are simultaneously degeneration (they linearize and thermally stabilize the FETs) and your **bias ammeters**: V = I·R, so 1.40 A reads as 0.658 V. You will set bias by trimming for that voltage.
- **DC feedback** (R_fb 82 k from OUT_NODE to Q1's gate, against R_in 47 k to ground) holds OUT_NODE near half-rail: if the output drifts up, Q1's gate is pulled up, Q1 conducts harder, and the output comes back down. The same network applies *light* AC feedback — enough to set damping, deliberately not enough to scrub the 2nd harmonic. 📖 Pass, *Zen Amp* (the same self-biasing idea with one FET).
- **C_out (10 mF)** couples the speaker because a single-rail amp's output idles at +11 V. Corner ≈ 2 Hz into 8 Ω. R_dc bleeds the speaker side to 0 V. The Zobel (R_z + C_z) keeps the amp loaded at HF where speaker cables turn weird. 📖 Self, *Audio Power Amplifier Design*, output networks chapter.

2.2 The power supply: quiet beats stiff

Fig. 3 — Power Supply (one per stereo amp) · rev A — Darlington capacitance multiplier

120 V / 60 Hz build · delivers +22.3 V at 2.8 A to both channels · ripple ≈1.2 V pp at the reservoir → <5 mV pp at the rail



① **Darlington fix (ECR-1):** a lone MJE15032 at 2.8 A needs ~56 mA of base current, which would drop ~12 V across R_cm and collapse the rail. BD139 in front raises composite β to ~5000, so the drop across R_cm is <0.2 V. R_be gives the pass pair a clean turn-off path. Total drop ≈ 2×V_BE ≈ 1.6 V.
② R_cm · C_cm = 220 Ω · 2200 μF → τ ≈ 0.5 s: the rail rises softly over ~1 s at power-on (gentle on the speaker) and 120 Hz ripple is attenuated ≈ 51 dB at the base node.
③ D_p clamps Q_cm at power-off, when the rail-side capacitors would otherwise reverse-bias it while the reservoir discharges. ④ CL-60 tames the 225 VA toroid + 20 mF inrush surge (ECR-1).
C_res stores ≈6 J at 24 V. After power-off: wait 5 minutes, then VERIFY < 2 V across C_res with a DMM before touching anything.

SONUS MINIMAL — SE Class A 5 W	Fig. 3 — Power supply schematic, Darlington capacitance multiplier (rev A) Supersedes 03_power_supply.png (single-BJT multiplier — see ECR-1)	Scale: none · Sheet 1/1	2026-07-03 · LABBOOK §4

A single-ended stage has almost no power-supply rejection: whatever ripple is on the rail walks straight into the speaker. So the design spends its effort on *quiet*: toroid → bridge → 20 mF reservoir (ripple ≈ 1.2 V pp) → **capacitance multiplier**.

The multiplier is an emitter follower whose base sits on an RC-filtered copy of the raw rail (R_cm·C_cm, τ ≈ 0.5 s → ripple down ~51 dB). The emitter — your clean rail — follows the smooth base, minus ~1.6 V. It is not a regulator: the rail sags gently with mains voltage, which Class A does not care about, and it costs two transistors, not a control loop. rev 4 made it a **Darlington** (BD139 driving MJE15032) because the single-BJT version could not supply 2.8 A of load through a 220 Ω base feed — the arithmetic is in ECR-1. A CL-60 inrush thermistor and a reverse-clamp diode complete it. 📖 Elliott (ESP) Project 15; AoE §9.x (linear supplies).

2.3 Why Class A single-ended runs this hot

Bias must exceed the largest current you will ever ask the speaker for: 1.4 A standing gives ±1.4 A of swing, i.e. ~7 W into 8 Ω before the top device starves. Both FETs dissipate `(V_ds × 1.4 A)` continuously — ≈ 14 W each — and the PSU pass transistor another ~5 W. Heat is the design's dominant mechanical constraint; that is why Phase D is a whole phase. 📖 Self, Class-A chapter; Elliott, heatsink design article.

2.4 Reading list

Ref	What to read	Why
Nelson Pass, **"The Amp Camp Amp"** (AudioXpress 2012; firstwatt.com → articles; diyAudio ACA build guide)	whole article	This is the direct ancestor: same stack, same bootstrap trick
Nelson Pass, **"Zen Amp"** + **"Zen Variations"** (passdiy.com)	Zen; ZV1–2	Self-biasing feedback, SE Class A philosophy, matching
Horowitz & Hill, **The Art of Electronics** 3rd ed.	§3.1–3.2 (FETs), §2.2.5 & 2.4 (current sources, bootstrap), Ch. 9 (supplies & thermal)	The *why* under every step in this book
Douglas Self, **Audio Power Amplifier Design** 6th ed.	Class-A chapter; grounding chapter	Rigor on THD mechanisms, star grounding
Rod Elliott (sound-au.com), **Project 15** (capacitance multiplier), **"Heatsink design"**, **"Death of Zen (DoZ)"**	all three	Practical multiplier + thermal design + another SE Class A build narrative
Datasheets: **IRFP150N** (Infineon/IR), **MJE15032** (onsemi), **BD139**, **CL-60** (Amphenol)	transfer & SOA curves	Source of every Vgs/β number used in §3
REW (roomeqwizard.com) or ARTA manual	soundcard THD measurement setup	Phase F instrumentation

§3 Engineering Change Record

The rev 3 handoff listed eight open items. Each is resolved below **with the arithmetic shown** — check it, don't trust it. The result is BOM **rev 4** (`docs/BOM-rev4.csv`). Two of these (ECR-1, ECR-2) were build-blocking: the amp as drawn in rev 3 would not have met its spec.

ECR-1 — Cap multiplier would have collapsed *(open item 1 — blocking)*

Problem. MJE15032 at $I_C = 2.8$ A has $\beta \approx 50$–100 (datasheet hFE curves). Base current $I_B = 2.8 / 50 \approx$ **56 mA** worst-case; drop across $R_{cm} = 220\ \Omega \rightarrow$ **12.3 V**. The "+24 V" rail becomes ~11 V. Even at $\beta = 100$ you lose 6 V.

Fix. Darlington: BD139 ($\beta \approx 100$ at 30 mA) drives the MJE15032. Composite $\beta \approx 5000 \rightarrow$ I through $R_{cm} \approx 0.5$ mA $\rightarrow$ drop ≈ 0.1 V. Costs one extra V_{BE} (total drop ≈ 1.6 V), one TO-126 transistor, and R_be 2.2 k for clean turn-off. Added while in there: **D_p** (1N5408) clamps the pass pair when the rail-side capacitance holds voltage after power-off, and **RT1** (CL-60) tames the inrush of a 225 VA toroid charging 20 mF. *BOM: +BD139, +R_be, +1N5408, +CL-60.*

ECR-2 — IRFP150N shifts the operating point *(open items 2, plus stability — blocking)*

Problem. The circuit self-biases: with matched FETs the equilibrium is `Vg1 = Vout·R_in/(R_in+R_fb)` and `Vg1 = V_GS(1.4 A) + 0.66 V`. The design was narrated around the IRFP240 ($V_{GS} \approx 4.5$ V at 1.4 A) but the rev 3 BOM ships the **IRFP150N** ($V_{GS} \approx 3.4$ V warm). With the old 47 k : 47 k network, OUT_NODE self-sets at $2 \times (3.4 + 0.66) \approx$ **8.1 V** — not ~11 V. Down-swing dies at ≈ -6.5 V pk → **≈ 2.7 W** before clipping. Spec busted.

Fix (verified numerically — see `spice/` and the solver run recorded below).

- **R_fb 47 k → 82 k**: divider ratio becomes 47/129 = 0.364, so OUT_NODE ≈ (V_{GS}+0.66)/0.364 ≈ 11 V for a typical part. Input impedance stays 47 k; feedback gets slightly lighter (in the spirit of the original §4).
- **Rb_bot 10 k → 22 k default, selected per Table 4.3** (18 k / 22 k / 33 k against the V_{GS} you measure in Phase A), divider returned to OUT_NODE. Numeric check with a square-law IRFP150N model (Kp=25, Vto=3.0): trimmer lands 1.40 A at Rrv ≈ 3.4 k — mid-range — with OUT_NODE = 10.9 V; the select table trims 1.40 A across Vto 2.6–3.4 with OUT_NODE between 9.8 and 12.0 V. All acceptable (swing costs < 0.5 dB at the extremes).
- **Stability**: IRFP150N has higher gm and C_{iss} than the IRFP240. Gate stoppers stay (220 Ω / 100 Ω → gate pole ≈ 380 kHz); a **Zobel** (10 Ω + 100 nF) is added at OUT_NODE so the amp never sees an open HF load. Oscillation check is mandatory in Phase F.

ECR-3 — Reservoir sizing confirmed *(open item 3)*

$\Delta V = I \cdot t / C = 2.8\ \text{A} \times 8.3\ \text{ms} / 20\ \text{mF} \approx$ **1.2 V pp**. Trough ≈ 24.2 – 0.6 ≈ 23.6 V. Multiplier needs rail + $V_{CE(sat)}$+margin ≈ 22.3 + 1.0 = 23.3 V < 23.6 V. ✅ Margin is real but thin — which is exactly what a cap multiplier wants (drop as little as possible, always stay under the trough). No change.

ECR-4 — The honest rail is ≈ 22.3 V, not 24 V *(open items 3/4 context)*

18 VAC nominal → at ~22 % load the Hammond delivers ≈ 18.6 VAC → peak 26.3 V – 1.8 V bridge – 0.6 V half-ripple – 1.6 V Darlington ≈ **22.3 V**. Consequences: OUT_NODE ≈ 11 V, max swing ≈ ±8.8 V pk → **≈ 4.8–5.5 W** at clip. Target met; all tables in this book use real numbers. Output corner: 10 mF into 8 Ω = 2.0 Hz ✅ (open item 4 closed, C_out sees ~11 V DC on a 50 V part ✅).

ECR-5 — Heatsink locked at 10″ per channel *(open item 5)*

Real dissipation is 22.3 V × 1.4 A ≈ **31 W per channel** (14.4 W per FET). At 0.5 °C/W the sink rises 15.6 °C → ≈ 41 °C in a 25 °C room; junctions ≈ 65 °C. The handoff's "~8 in" of HeatsinkUSA 10.080″ profile is marginal at natural convection; **10″ per channel** buys real margin for ~$8. Acceptance is *measured*, not asserted: Phase F thermal soak requires sink ≤ 50 °C after 1 h (≈ 0.8 °C/W absolute ceiling; ≤ 45 °C is the target).

ECR-6 — Unverified electrolytics locked *(open item 6)*

C_cm → **Nichicon UPW1H222MHD** (2200 µF 50 V), Cb_filt → **UPW1H101MPD** (100 µF 50 V); alternates Panasonic EEU-FC. Both 50 V to match the rest of the build. (The rev 3 Panasonic ECA 35 V placeholders were never stock-verified.) Confirm stock at cart, as with everything.

ECR-7 — Mechanical fit *(open item 7)*

The 4880SG insulator kit is TO-247-sized — right for Q1/Q2, wrong for the TO-220 Q_cm. BOM adds a TO-220 mica + bushing kit (verify the exact MPN at cart; any TO-220 kit works). RCA footprint and binding-post holes get checked against the chassis in Phase D, step D-7.

ECR-8 — Rectifier noise *(open item 8)*

Standard silicon bridge stays; the multiplier's ~51 dB of ripple attenuation is the noise strategy. If (and only if) Phase F shows 120 Hz-harmonic hash riding on the output, retrofit 0.1 µF + 10 Ω snubbers across each bridge leg or swap to the Schottky alternate. Decision deferred to measurement, where it belongs.

§4 The Final Design

These four artifacts are the single source of truth. When any other document disagrees, these win:

Artifact	File
Channel schematic (rev A)	`diagrams/fig02-channel-schematic.svg`
PSU schematic (rev A)	`diagrams/fig03-psu-schematic.svg`
BOM rev 4	`docs/BOM-rev4.csv`
SPICE decks	`spice/sonus-minimal-channel.cir`, `spice/sonus-minimal-psu.cir`

4.1 DC voltage map (no signal, warm, per channel)

Measure everything against the star ground (black DMM lead on the star bolt).

Test point	Node	Expected	Acceptable	Symptom if outside
PSU	reservoir RAW+	+24.2 V	23–26 V	transformer/bridge problem
PSU	multiplier base	+24.1 V	raw − 0.3 V	R_cm open / C_cm shorted
TP1	+RAIL	**+22.3 V**	21–23.5 V	Darlington fault (see ECR-1)
TP2	VBIAS2	≈ +15.0 V	13.5–16.5 V	divider / bootstrap fault
—	Q2 source	OUT + 0.66 V	—	R_s2 open if far off
TP3	OUT_NODE	**≈ +11 V**	9.8–12.3 V	see Table 4.3 / re-select Rb_bot
—	Q1 gate node (G1)	0.364 × OUT ≈ +4.0 V	3.6–4.5 V	R_fb/R_in fault
TP4	Q1 source (across R_s1)	**0.658 V = 1.40 A**	±0.010 V after warm trim	that's the bias — trim it
—	Speaker + (after C_out)	0 V	±20 mV	R_dc open / C_out leaking

4.2 Where *your* OUT_NODE will sit

OUT_NODE is not independently trimmable — it self-sets at `(V_GS + 0.66 V) / 0.364` once bias is trimmed to 1.40 A. Low-threshold FET pairs land near 10 V, high-threshold pairs near 12 V. Anything in 9.8–12.3 V is **correct**, costs at most ~0.5 dB of one-sided headroom, and must simply be *recorded*, not "fixed".

4.3 Rb_bot selection table (used in Phase C, from Phase A data)

Using the pair's V_GS measured in the Fig. 4 jig (≈1.25 A):

Measured V_GS (jig)	Stuff Rb_bot	Expected OUT_NODE
< 3.15 V	18 k	≈ 9.8–10.5 V
3.15 – 3.55 V	**22 k (default)**	≈ 10.4–11.7 V
3.55 – 3.90 V	33 k	≈ 11.8–12.3 V
> 3.90 V (rare)	39 k (order if needed)	≈ 12.5 V

Final authority is the trimmer, not the table: if V(R_s1) at **max** trim is still < 0.658 V → power down, stuff the next **higher** Rb_bot. If at **min** trim it already exceeds 0.658 V → next **lower** value.

4.4 BOM rev 4 — deltas from rev 3

Full list: `docs/BOM-rev4.csv` (DigiKey-import compatible, same columns as rev 3).

Change	What	Why
NEW	BD139 (Q_d), R_be 2.2 k, 1N5408 (D_p), CL-60 (RT1)	ECR-1 Darlington multiplier + inrush
NEW	R_z 10 Ω 2 W, C_z 100 nF (0.1 μF line qty 6→12)	ECR-2 Zobel
CHANGED	R_fb 47 k → **82 k** (47 k line qty 18→12)	ECR-2 re-centre OUT_NODE
CHANGED	Rb_bot 10 k → **22 k** + select kit 18 k/33 k	ECR-2 bias trim range
CHANGED	IRFP150N order qty 12 → **20**	Vgs sorting headroom (12 used)
LOCKED	C_cm → UPW1H222MHD, Cb_filt → UPW1H101MPD (both 50 V)	ECR-6
CHANGED	4880SG qty 15 → 12; **+ TO-220 kit ×3** for Q_cm	ECR-7
CHANGED	HS1: cut length ~8″ → **10″ per channel**	ECR-5
TOOLS	6.8 Ω 10 W (jig), 2 × 8 Ω 50 W (dummy), thermal compound	bench, not per-amp

Ordering notes (inherited from rev 3, still true): `0.47AECT-ND` imports in the *DigiKey P/N* field, not MPN · C_out/C_res are one merged line, qty 12 · T1 and HS1 will not auto-match (Hammond is stocked; HeatsinkUSA is direct-order) · stock always re-verified at the cart.

§5 Safety, Tools, Workspace

⚠ The three ways this project can hurt you

1. **Mains (lethal).** Everything left of the transformer secondaries in Fig. 3 is 120 V. Rules, no exceptions: work unplugged unless a step says otherwise; heat-shrink every mains joint; the IEC earth pin bonds to bare chassis metal with its own bolt *first, before any other wiring exists*; never defeat the earth; never float the fuse.
2. **Stored energy.** 20 mF at 24 V is ~6 J — enough to weld a screwdriver tip and destroy whatever it splashes. ⏱ After every power-down: **wait 5 minutes, then verify < 2 V across C_res with your DMM** before touching. Every time. R_bleed does the work; the meter is proof.
3. **Heat.** Sinks run 45–50 °C (uncomfortable), R_s resistors and Q_cm hotter. Fingers off until measured.

Tools (bench minimum)

Tool	Used for	Notes
DMM reading to 1 mV	everything	two are better (bias + voltage at once)
Oscilloscope ≥ 20 MHz	ripple, oscillation, square waves	RIGOL-class is plenty
Dim-bulb tester (Fig. 8)	every first power-up	build it in Phase B — mandatory
Bench PSU 12 V ≥ 1.5 A (or wall adapter)	Fig. 4 matching jig	
Signal source	1 kHz sine / squares	soundcard + REW is fine
Dummy loads 2 × 8 Ω ≥ 50 W	Phases B/E/F	never test into speakers first
Soldering iron ≥ 60 W + large tip	bus wire, snap-in caps, lugs	small irons fail on the big joints
IR thermometer or TC probe	Phase F thermal soak	±2 °C is fine
Hand tools	drill + bits, taps M3 (or #4-40), crimper, heat gun	Phase D

Workspace: non-conductive bench mat, good light, no carpet-shuffle ESD (MOSFET gates die silently — handle by the package, touch the chassis first), a printed Fig. 2/3/5 within reach.

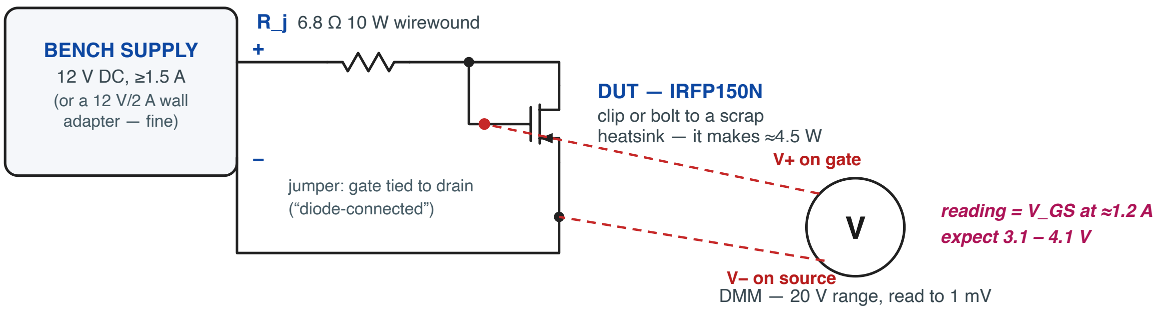
THE BUILD

Phase A — Inventory & MOSFET Matching

Goal: every part verified against BOM rev 4; twenty IRFP150Ns measured and sorted into six matched pairs (plus spares); Rb_bot values selected per Table 4.3. 📖 Pass on matching (Zen/ACA).

Fig. 4 — V_GS Matching Jig (Phase A)

Diode-connected MOSFET at ≈1.2 A — right at the amp's operating point. One resistor, one supply, one meter.



Why 6.8 Ω:
I = (12 V – V_GS) / 6.8 Ω ≈ (12 – 3.5)/6.8 ≈ 1.25 A ≈ the amp's 1.4 A bias point, so the ranking you measure here is the ranking that matters in-circuit.

Per device (≈2 min each):
1. Power OFF. Clip DUT to the heatsink, hook up G-D jumper + leads.
2. Power ON. The resistor and DUT get hot — that is expected.
3. Wait 60 s for thermal settle, then read V_GS to the millivolt.
4. Log in Table A.2 with the device's marker number. Power OFF.
5. Sort all devices by V_GS. Pair ADJACENT values.

Accept a pair when ΔV_GS ≤ 25 mV. Q1+Q2 of one channel = one pair.
Same room temp for every reading; handle by the plastic, not the tab.

SONUS MINIMAL	Fig. 4 — V_GS matching jig (rev A)	2026-07-03 · LABBOOK Phase A

A-1 Inventory

- ☐ Check every BOM line against the delivery. Record shortfalls: ______________________
- ☐ Verify the insulator kits: 12 × TO-247 mica + shoulder washers, 3 × TO-220 kits (ECR-7).
- ☐ Verify T1 label: **1182G18, dual 115 V primary, dual 18 V secondary**.
- ☐ Number each IRFP150N **1–20** with a paint marker on the plastic (never the tab).

A-2 Build the jig (Fig. 4)

- ☐ 12 V supply → 6.8 Ω 10 W → DUT drain; gate jumpered to drain; source → supply return.
- ☐ DMM across gate–source. DUT clipped/bolted to any scrap heatsink (it makes ~4.5 W).
- ☐ Sanity check: I = (12 – V_GS)/6.8 ≈ 1.2 A — right at the amp's operating point, which is why the ranking measured here is the ranking that matters.

A-3 Measure all 20 (≈ 2 min each)

Protocol per device: power off → mount DUT → power on → ⏱ wait 60 s → read V_GS to 1 mV → power off → next. Same room temperature throughout; if interrupted > 30 min, re-measure one earlier device as a drift check.

Table A.2 — V_GS log (also in Appendix A for units 2, 3)

#	V_GS (V)	#	V_GS (V)	#	V_GS (V)	#	V_GS (V)
1		6		11		16	
2		7		12		17	
3		8		13		18	
4		9		14		19	
5		10		15		20	

A-4 Sort and pair

- ☐ Re-write the list sorted by V_GS. Pair **adjacent** values: pair = (Q1, Q2) of one channel.
- ☐ Accept pairs with ΔV_GS ≤ **25 mV**. Six pairs needed; best-matched pairs go to unit SM-001.
- ☐ For each pair, choose Rb_bot from **Table 4.3** and write it here:

Unit / channel	Devices #,#	Pair V_GS	ΔV_GS (mV)	Rb_bot selected
SM-001 L				
SM-001 R				
SM-002 L				
SM-002 R				
SM-003 L				
SM-003 R				

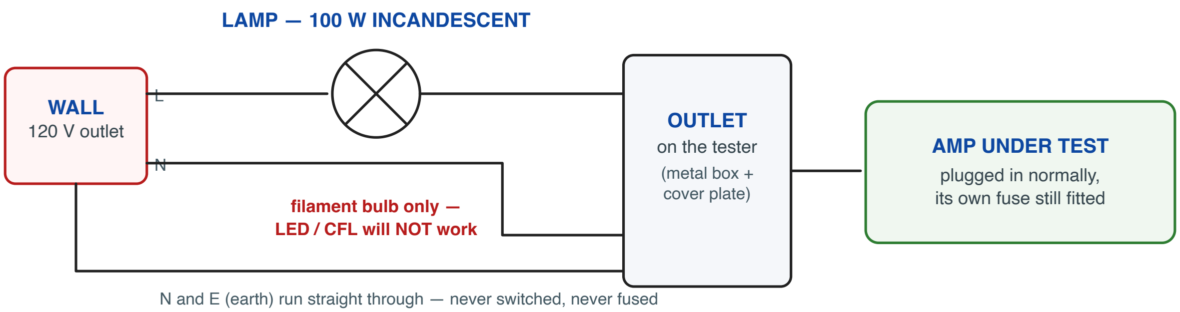
GATE A: ☐ all six pairs ≤ 25 mV ☐ Rb_bot chosen per pair ☐ leftover devices bagged as labeled spares. *(A high-V_GS straggler pair > 3.90 V? Order 39 k before Phase C.)*

Phase B — Power Supply Build & Solo Test

Goal: one PSU built and proven alone: +22.3 V clean rail into a dummy load, ripple measured, before any amplifier sees it. **References:** Fig. 3 (schematic), Fig. 6 (board), Fig. 8 (dim bulb), Fig. 10 (milled option). ⚠ This phase contains the mains work. Re-read §5 first.

Fig. 8 — Dim-Bulb Tester (every first power-up goes through this)

A filament bulb in series with the LINE limits any fault to a warm glow instead of a bang. Build one or buy one — do not skip it.



What the bulb tells you (this amp draws ≈95 W when healthy):

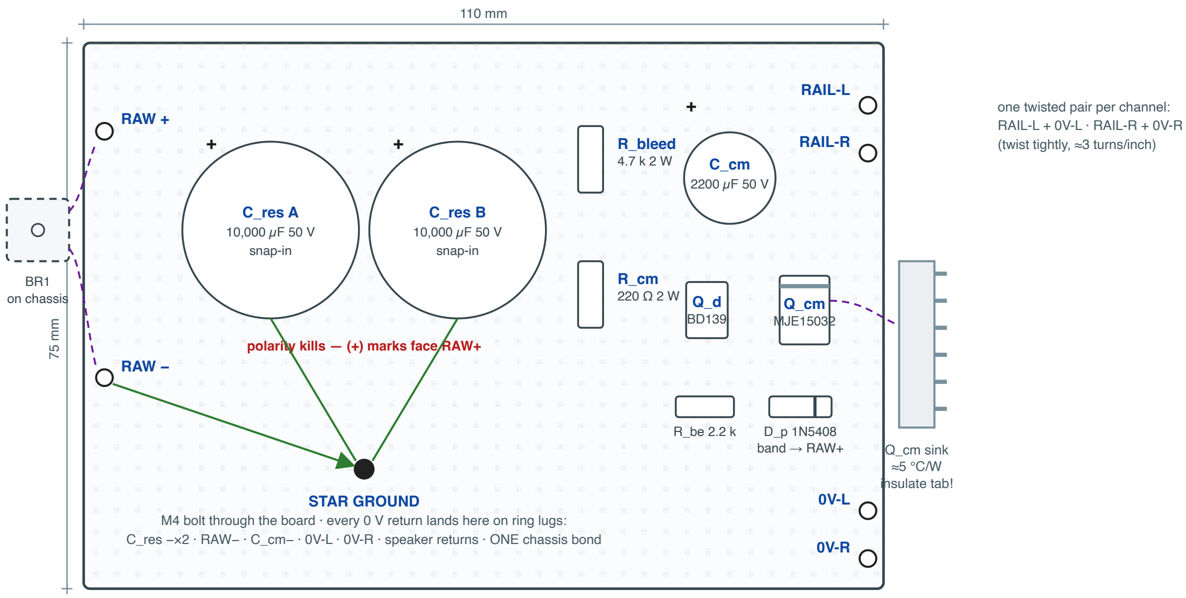
- **Bright flash, then settles to a steady half-glow:** NORMAL — inrush charged the reservoir, amp is drawing bias. Proceed.
- **Full brightness, steady:** FAULT — a short (miswired bridge, reversed cap, tab touching heatsink). Power off, discharge, find it.
- **Dark, and no rail voltage:** open circuit — check F1, IEC wiring, transformer connections.

Note: with the bulb in series the rail will sit LOW (the bulb steals volts) — that is fine for smoke-testing. Do final bias trim WITHOUT the bulb.

SONUS MINIMAL	Fig. 8 — Dim-bulb tester (rev A)	2026-07-03 · LABBOOK Phase B/E

Fig. 6 — PSU Board, perfboard placement map (one per amp)

110 × 75 mm, 0.1″ perfboard. The bridge rectifier is NOT on this board — it bolts to the chassis floor. Connectivity from Fig. 3.



Rules: ① snap-in caps: enlarge holes to 2 mm, solder short and stiff ② the 2.8 A paths (RAW+ → caps → Q_cm collector, emitter → RAIL pads) are bus wire ③ R_bleed spans RAW+ to star — mount it 2 mm proud, it dissipates ≈0.12 W continuously ④ keep C_cm right next to Q_d's base pin ⑤ nothing else shares the star bolt's ring-lug stack order: lug stack from board up: C_res returns → RAW– → channel 0 V → speaker returns → chassis bond lug LAST (top)

SONUS MINIMAL	Fig. 6 — PSU board placement (perfboard) · rev A	2026-07-03 · LABBOOK Phase B

B-1 Build the dim-bulb tester (Fig. 8) — once, keep forever

- ☐ 100 W **incandescent** bulb in series with LINE only; N and E straight through; metal box, earthed. ☐ Test it: a lamp plugged into the tester lights via the bulb.

B-2 Stuff the PSU board (Fig. 6 or Fig. 10)

Order: small parts → transistors → big caps last.

- ☐ R_cm, R_be, R_bleed (2 mm proud), D_p (**band → RAW+**), C_cm (polarity!).
- ☐ BD139 + MJE15032: **pinouts differ** (BD139 = E-C-B, MJE = B-C-E facing the label). Triple-check against Fig. 6/10 before soldering.
- ☐ Q_cm on its ~5 °C/W bracket with TO-220 mica kit. **Buzz tab-to-bracket: open circuit.** Record: tab↔sink R = ______ (must be OL)
- ☐ C_res ×2 (snap-in, polarity, fully seated), star bolt fitted with lug stack per Fig. 6.
- ☐ Continuity sweep against Fig. 3, every adjacent island/net pair: no unexpected beeps.

B-3 Mains side (chassis may be loose-assembled for this)

- ☐ **First wire in the box:** IEC earth → bare-metal chassis bolt, star washer, ring lug. Buzz: IEC earth pin ↔ any chassis point < 0.5 Ω. Record: ______ Ω

- ☐ IEC (switched, fused, F1 = 2 A slow fitted) → RT1 (CL-60) in LINE → T1 primaries **in parallel** (120 V): join blk–blk/red–red per Hammond datasheet — verify colors against the datasheet, not memory. *(230 V: primaries in series, F1 = 1 A slow.)*
- ☐ Heat-shrink every mains joint. Tug-test each one.

B-4 Secondary phasing test ⚠ (secondaries NOT yet paralleled)

1. ☐ Temporarily join ONE end of secondary A to ONE end of secondary B.
2. ☐ Power up through the dim bulb. Measure AC across the two free ends: **V = ______ VAC** → ≈ 0 V: phased correctly — the free ends may be joined. → ≈ 36–40 V: flip one winding's leads. Never join ends that read 36 V.
3. ☐ Power off, ⏱ discharge check, make both parallel joints permanent.

B-5 First light (PSU alone, dummy load)

- ☐ Bridge bolted to chassis (thermal compound), AC → bridge, bridge → board, **8 Ω 50 W dummy from RAIL to 0 V** (≈ 2.8 A — the real load current).
- ☐ Power up **through the dim bulb**: flash → half-glow = normal. Record behavior: __________
- ☐ Power off, discharge, inspect. Then power up **without** the bulb and record, in order:

Measurement	Expected	Unit 1	Unit 2	Unit 3
V(RAW+) DC	≈ 24.2 V			
Ripple at RAW+ (AC, scope)	≈ 1.2 V pp @ 120 Hz			
V(base of Q_d)	RAW − 0.15 V			
V(RAIL) DC	**21.5–23 V**			
Ripple at RAIL	**< 5 mV pp**			
Q_cm sink temp after 15 min	< 55 °C			
Power-off: V(C_res) after 5 min	< 2 V			

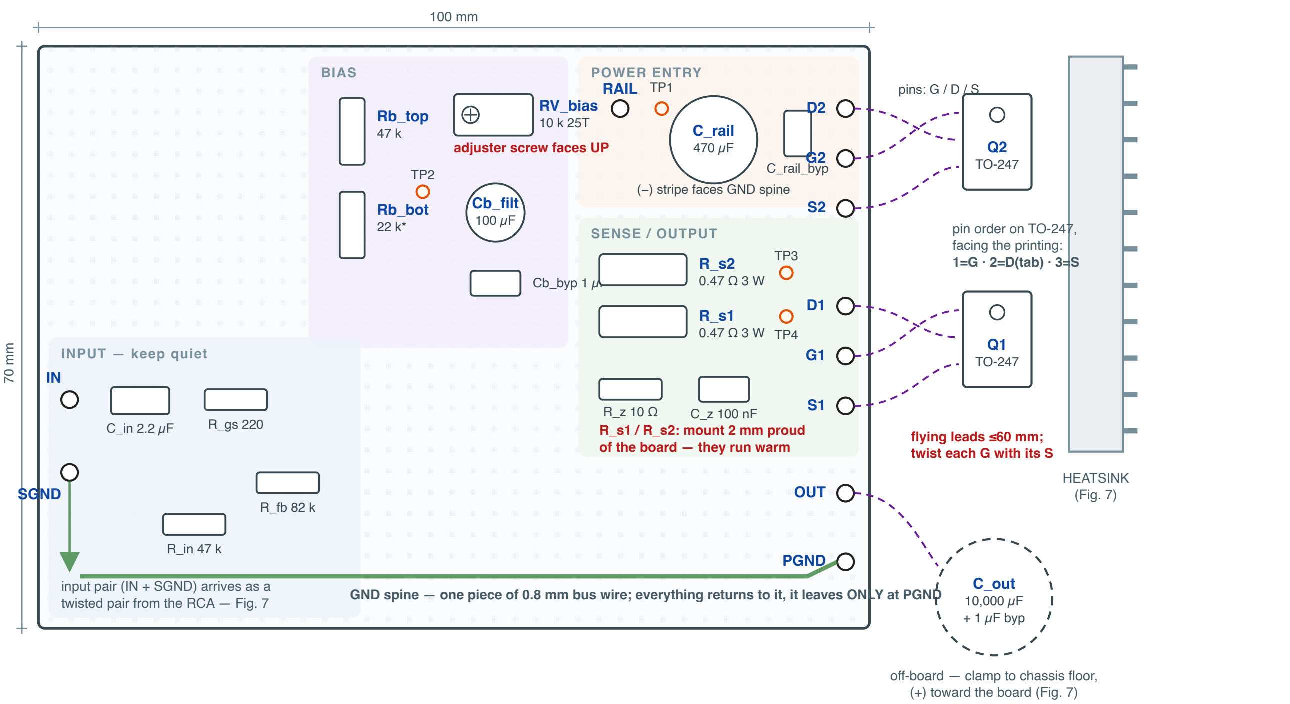
GATE B: □ all rows in range □ phasing test recorded □ earth bond < 0.5 Ω □ tab isolation OL. A rail that reads ~raw−12 V means the Darlington is miswired (that's the exact ECR-1 failure signature — check BD139 pinout first).

Phase C — Channel Boards (×2 per amp)

Goal: two stuffed, buzzed-out channel boards with matched FET pairs on flying leads. **References:** Fig. 2 (schematic — print it), Fig. 5 (perfboard) or Fig. 9 (milled).

Fig. 5 — Channel Board, perfboard placement map (build two per amp)

100 × 70 mm, 0.1″ perfboard (BusBoard PAD-1 or similar). Placement + zones here; connectivity comes from Fig. 2. Wire point-to-point with bus wire underneath.



Rules: ① the 1.4 A path (RAIL→D2 · S2→OUT · D1→OUT · S1→spine) is bus wire, never thin hookup wire ② Rb_bot* is selected in Phase A (Table 4.3) ③ Cb_filt (+) faces TP2, its (–) goes to OUT — not to the spine ④ nothing from the INPUT zone touches the spine except at the single SGND→spine tie shown ⑤ leave TP1–TP4 as wire loops you can clip a probe to ⑥ R_fb runs OUT→input zone along the board edge

SONUS MINIMAL	Fig. 5 — Channel board placement (perfboard) · rev A	2026-07-03 · LABBOOK Phase C

Per board:

- ☐ **C-1** Lay the GND spine (0.8 mm bus wire) and the 1.4 A path (RAIL→D2, S2→OUT, D1→OUT, S1→spine) in bus wire first — power wiring before signal wiring.
- ☐ **C-2** Resistors: R_in 47 k, R_fb **82 k**, R_gs 220, R_g2 100, Rb_top 47 k, **Rb_bot = value selected in Table A/4.3 for THIS pair** = ______ , R_dc 100/2 W, R_z 10/2 W, R_s1 & R_s2 0.47/3 W (both 2 mm proud).
- ☐ **C-3** RV_bias as rheostat (wiper + one end), adjuster accessible. **Preset to minimum:** ohmmeter across VBIAS2→junction reads ≈ 0 Ω contribution (25 turns CCW). Record preset resistance VBIAS2↔OUT path: ______ (should ≈ Rb_bot alone).
- ☐ **C-4** Capacitors: C_in film, C_rail + 100 nF at the rail entry, **Cb_filt (+ toward VBIAS2) and Cb_byp between VBIAS2 and OUT_NODE — NOT to ground** (§2.1 — the classic mistake), C_z, TP1–TP4 wire loops.
- ☐ **C-5** FET flying leads ≤ 60 mm, gate twisted with its source return; leads labeled G1/D1/S1/G2/D2/S2. TO-247 pin order facing the label: **1=G, 2=D(tab), 3=S**.

- ▢ **C-6** Buzz-out against Fig. 2 — check every net, then check three *non*-connections: VBIAS2↔GND (must be ≥ 32 k through resistors, not 0), OUT↔GND (≥ ~100 Ω via R_dc path), RAIL↔OUT (open at DC path except through FETs — with no FETs fitted: ≥ 47 k + divider).

Record (per board): built by ______ date ______ · Rb_bot fitted ______ · RV preset verified □ · buzz-out clean □

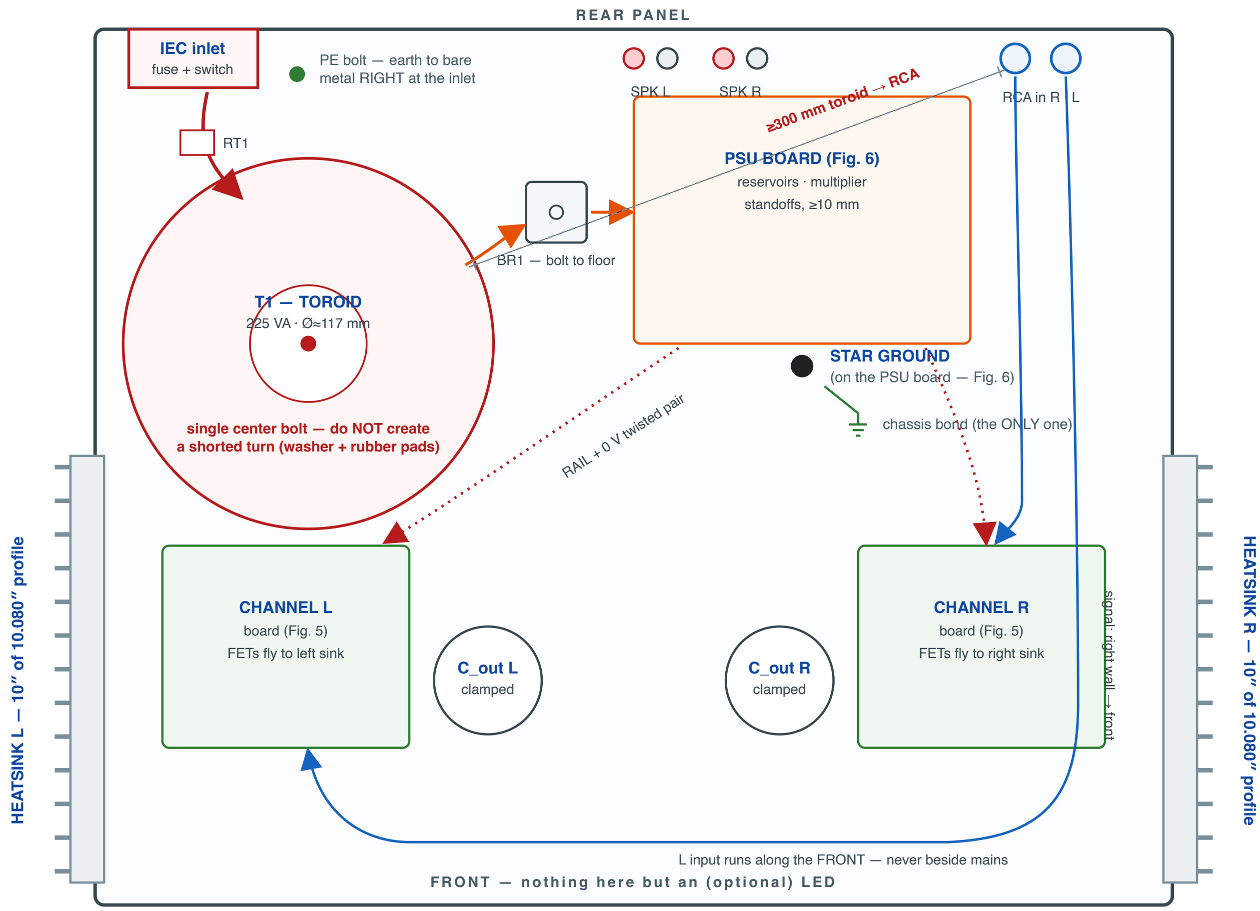
GATE C: □ both boards complete □ Cb caps verified bootstrapped to OUT □ RV at minimum □ FET pairs still bagged with their board (don't mix pairs between channels!).

Phase D — Heatsinks & Mechanical

Goal: FETs mounted, insulated, verified; chassis drilled; everything placed per Fig. 7.

Fig. 7 — Chassis Layout, top view (lid off, rear at top)

≈320 × 260 × 80 mm plate/box chassis. The single organizing idea: mains lives rear-left, signal lives right+front, and they never run side by side.



① heatsinks are the side walls: fins OUTSIDE, vertical, free air above and below (≥10 mm feet) ② speaker wiring (posts→boards) may take any route — it is low-impedance and robust
③ every wire that crosses the mains zone is either mains itself or absent ④ C_out caps clamp to the floor beside their boards; keep them off the heatsinks (electrolytics hate heat)

SONUS MINIMAL · Fig. 7 — Chassis layout, top view · rev A · 2026-07-03 · LABBOOK Phase D/G

- ☐ **D-1** Cut/receive 2 × 10″ of 10.080″ profile per amp. Deburr.
- ☐ **D-2** Drill + tap M3 (or #4-40) for each TO-247, positions per Fig. 7 (FETs land behind their board). Flat-file any drilling burrs — burrs puncture mica.
- ☐ **D-3** Mount each FET: thin thermal compound *both* sides of mica, shoulder washer in the package hole, screw snug + ¼ turn (over-torque cracks mica). 📖 Elliott, heatsink article.
- ☐ **D-4 The isolation test (never skip):** DMM ohms, tab ↔ heatsink, every device:

Device	Unit 1	Unit 2	Unit 3
Q1 L / Q2 L	OL / OL		
Q1 R / Q2 R	OL / OL		

Device	Unit 1	Unit 2	Unit 3
Q_cm	OL		

Anything < OL: remount with fresh mica. The tabs carry OUT_NODE and +RAIL — a shorted tab is a dead channel and possibly a dead speaker.

- ☐ **D-5** Chassis: place per Fig. 7 (toroid rear-left near IEC, PSU center, channel boards at their sinks, C_out clamped beside each board, RCA rear-right). Toroid center bolt with rubber pads — **one** bolt, no conductive loop over the toroid (shorted turn).
- ☐ **D-6** Ventilation: ≥ 10 mm feet; if the amp gets a lid, slot it above the sinks.
- ☐ **D-7** (ECR-7) Verify RCA footprint and binding-post hole sizes against the drilled panel *before* final mounting. RCA jacks are insulated from the panel (signal ground goes to the star, not the chassis).

GATE D: □ isolation table all OL □ no shorted turn □ boards standoff-mounted □ Fig. 7 distances respected (toroid↔RCA ≥ 300 mm).

Phase E — First Power-Up & Bias

Goal: each channel biased to 1.40 A warm, DC map verified. **Setup:** PSU proven (Gate B) · one channel connected at a time · **8 Ω dummy load** on the output (never a speaker in this phase) · no input signal, RCA shorted with a shorting plug. ⚠ Dim bulb for every first power-up. Both DMMs out: one on TP4 (bias), one roving.

Per channel:

1. ☐ **E-1** RV_bias still at minimum (Gate C). Dummy load on. Scope on TP3 if available.
2. ☐ **E-2** Power up through the dim bulb: flash → dim. Steady bright = fault: off, discharge, Troubleshooting table.
3. ☐ **E-3** Bulb out, power direct. Read the cold map: TP1 ______ V · TP2 ______ V · TP3 ______ V · **TP4 ______ V.** TP4 should read *low* (under-biased) — that's the safe starting state. If TP4 already > 0.658 V at min trim → power off, stuff next-lower Rb_bot (§4.3).
4. ☐ **E-4** Trim up: turn RV_bias slowly (25-turn ≈ many turns — patience) watching TP4 rise to **0.658 V**. If it tops out below 0.658 → next-higher Rb_bot (§4.3).
5. ☐ ⏱ **E-5** Warm soak 30 min at bias. Heat makes V_GS fall → current rises → **re-trim to 0.658 V warm.** Repeat once more at 45 min; two consecutive readings within 5 mV = settled.
6. ☐ **E-6** Record the warm DC map vs §4.1 (tolerances there):

Channel	TP1 rail	TP2 VBIAS2	TP3 OUT	TP4 bias	SPKR+ DC	Sink °C
SM-__ L						
SM-__ R						

7. ☐ **E-7** Power-off behavior: rail collapses over ~2 s, no bangs into the dummy. ✓

GATE E: ☐ both channels 0.658 V ±10 mV warm ☐ TP3 within 9.8–12.3 V ☐ SPKR+ within ±20 mV ☐ nothing over 55 °C except as expected.

Phase F — Verification & Performance

Goal: prove stability, then character. Dummy loads still on. Instrument: scope + signal source (or soundcard + REW/ARTA through a 10:1 protection divider — see F-4 note).

- ☐ **F-1 Oscillation check (before anything else).** No input, scope on OUT at full bandwidth, 5 mV/div: any MHz fuzz = STOP. Remedy ladder: reseat gate leads (twisted? short?) → raise R_{gs} 220→470 → verify Zobel. Clean trace? Record noise floor: ______ mV pp.
- ☐ **F-2 Gain.** 1 kHz, 100 mV rms in → V_{out} = ______ rms → gain = ______ (expect ≈ 9–11×, 19–21 dB). Channels match within 0.5 dB: L ______ dB, R ______ dB.
- ☐ **F-3 Square waves.** 1 kHz and 10 kHz, ~1 V pk out: clean corners, no ringing (slight rounding fine, LF tilt at 1 kHz ≈ none — corner is 2 Hz). Photo/screenshot filed: □
- ☐ **F-4 THD spectrum — the design goal made measurable.** 1 kHz at ≈ 1 W (2.83 V rms) into 8 Ω dummy; FFT. **Acceptance: H2 dominant, each higher harmonic lower than the last, no rising high-order tail.** H2 = ______ dBc · H3 = ______ dBc · THD = ______ %. H3 above H2, or hash: check bias (crossover-like starvation = under-biased), oscillation, or too-hard clipping level. *(Soundcard inputs die above ~2 V — always use a divider.)*
- ☐ **F-5 Power at clip.** Raise level until visible flattening: V_{out} at clip onset ______ V pk → $P = V^2/2R$ = ______ W (expect 4.5–5.5 W). Note which side clips first (top starve vs bottom cutoff) — asymmetry is normal, record it.
- ☐ **F-6 Hum & noise.** Input shorted, output reading (true-rms mV or FFT): ______ µV (target < 100 µV; < 300 µV acceptable). 120 Hz spikes → grounding (Troubleshooting).
- ☐ **F-7 Thermal soak, 1 h at idle.** Log every 10 min:

t (min)	0	10	20	30	40	50	60
Sink L °C							
Sink R °C							
TP4 L (V)							
TP4 R (V)							

Acceptance (ECR-5): sinks ≤ 50 °C (≤ 45 °C target) at 25 °C ambient; bias drift < 5 % in the final half hour.

- ☐ **F-8** Repeat F-1 once more after the soak (hot oscillation check).

GATE F: □ no oscillation cold or hot □ H2-dominant spectrum □ ≥ 4.5 W at clip □ noise in range □ thermal acceptance met. *This gate is what "competition class" means: numbers, not vibes.*

Phase G — Final Assembly, Listening, Sign-off

- ▢ **G-1** Final chassis assembly per Fig. 7; dress mains rear-left, signal front/right; every wire crossing checked against the "never side by side" rule.
- ▢ **G-2** Full-assembly safety pass: earth bond from IEC pin to chassis ______ Ω (< 0.5); no mains conductor strain; fuse correct (2 A slow); discharge-check routine posted inside lid.
- ▢ **G-3** One more DC map (assembled state) — compare to Phase E table; investigate any shift > 5 %.
- ▢ **G-4 First music.** Power on, ⏱ 30 s settle, *then* connect speakers (≥ 92 dB/W/m — non-negotiable #1). Modest level first. Turn-on thump should be a soft "whump" or less (the ~1 s rail ramp is doing that for you); power-off with speakers connected is fine.
- ▢ **G-5** Listen: noise floor at the listening seat (silence between tracks), bass control, the 2nd-harmonic "roundness" on massed strings/voice. Notes: ______________________________
- ▢ **G-6** Fill the serial plate + Appendix A log; sign the cover table.

Congratulations — you built a measured, documented, competition-grade Class A amplifier. Units 2 and 3: repeat B→G (Phase A already covered all devices).

Troubleshooting

Symptom	Most likely	Check / fix
Dim bulb steady bright	mains short; reversed C_res; tab shorted to sink	isolation test D-4; bridge orientation; C_res polarity
Rail ≈ 11–14 V instead of 22	ECR-1 signature: Darlington miswired / BD139 pinout	BD139 = E-C-B! Rebuild per Fig. 6/10
Rail ≈ 20 V and Q_cm very hot	D_p reversed (conducting)	band faces RAW+
TP4 stuck near 0, trim does nothing	Rb_bot too big for pair; RV wired as pot not rheostat; Q2 gate open	§4.3 select; rewire wiper+end; buzz R_g2 path
TP4 pegged high at min trim	Rb_bot too small for pair	next-lower value per §4.3
Gain ≈ 0.3×, everything DC-correct	**Cb_filt/Cb_byp bypassed to GND, not OUT**	§2.1 — move them
OUT_NODE < 9.5 V or > 12.5 V (bias correct)	pair mismatched vs Rb_bot table	recheck Phase A pairing; §4.3
MHz fuzz on output	layout/gate leads/oscillation	F-1 remedy ladder
120 Hz buzz in speaker	ground loop; return not at star; input lead near toroid	one-wire-one-return audit vs Fig. 6; Fig. 7 routing
60 Hz hum, level-dependent	input cable / RCA insulation to chassis	insulated RCA? signal gnd at star only
Hiss audible on 95 dB+ speakers at seat	normal-ish; check noise number F-6	if > 300 μV: bias divider caps, C_cm health
One channel 1–2 dB quieter	gain part tolerance; bias off	re-measure F-2; verify 0.658 V both
Smell of hot resistor at idle	R_s or R_dc under-rated or bias high	verify 0.658 V; parts per BOM (3 W / 2 W)
Thump on power-off with cans (App. C)	cap discharge path	never power-cycle with headphones on

Appendix A — Per-unit measurement log

Duplicate of every record table (A.2, B-5, D-4, E-6, F-2..F-7) for units SM-002 and SM-003 — photocopy or reproduce in your notebook. A unit without its filled log has not completed Phase G by definition.

Appendix B — SPICE decks

`spice/sonus-minimal-channel.cir` and `spice/sonus-minimal-psu.cir` run in LTspice or ngspice (`ngspice -b file.cir`). Expected results are commented in each deck. Uses: sanity-check a proposed change (different Rb_bot, heavier feedback), see the trim range (sweep `Rrv` 1k–9k), watch PSU start-up. The MOSFET model is approximate (Vto = 3.0 typ) — your parts differ, which is the whole reason RV_bias exists. The rev-A operating point was verified numerically: trim lands 1.40 A at mid-range with OUT_NODE = 10.9 V (Vto = 3.0), and Table 4.3 selections hit 1.40 A across Vto 2.6–3.4.

Appendix C — Headphone output (optional, from the original design)

Unchanged from `docs/original/SE-ClassA-5W-amp.md` §11: an L-pad (470 Ω series / 47 Ω shunt, ½ W metal film) tapped from the speaker side of C_out into a switched ¼″ TRS (Neutrik NJ3FP6C or similar) whose NC contacts mute the speakers. –22 dB, source Z ≈ 43 Ω — right for HD600/650-class high-impedance cans, wrong for IEMs. **Plug in only after the supply settles; never power-cycle with headphones on.** Build it after Phase G, not before.

Appendix D — 230 V regions

F1 → 1 A slow (0218001.MXP) · T1 primaries in **series** per the Hammond datasheet · ripple period becomes 10 ms → reservoir ripple ≈ 1.4 V pp (still fine) · CL-60 stays. Everything else identical.

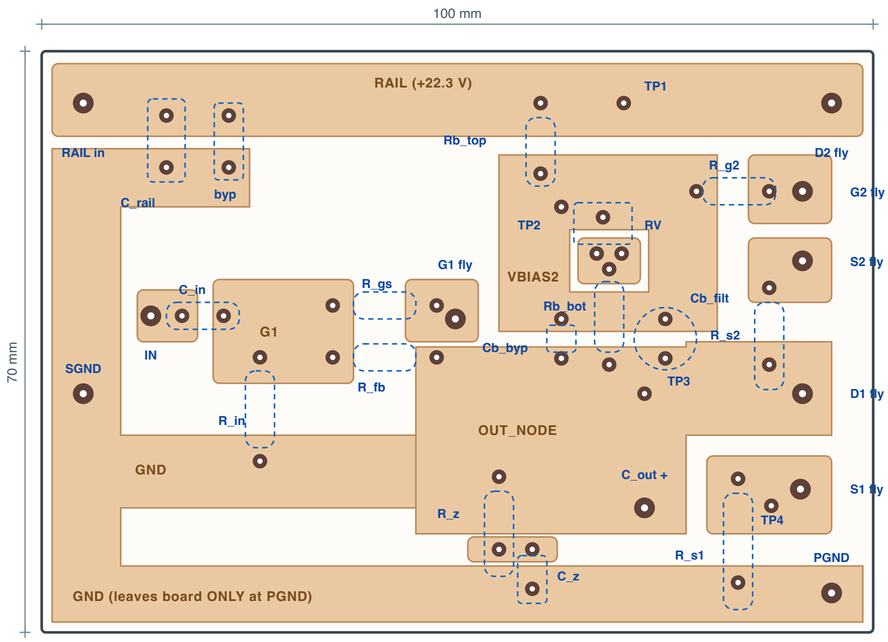
Appendix E — Isolation-milled boards (alternative to perfboard)

Fig. 9 (channel) and Fig. 10 (PSU) give single-sided island plans: net-true, placement-true, with real-ish pad pitches. They are plans, not Gerbers — redraw in your CAM at these positions, mirror for toolpaths, keep gaps ≥ 0.4 mm, and buzz out every island pair after milling. The 1.4 A / 2.8 A paths are wide pours; reinforce the marked Q_cm collector finger with bus wire. Everything else in Phases C/B applies unchanged.

Fig. 9 — Channel Board, isolation-milling plan (single-sided FR4)

Viewed from the COMPONENT side (x-ray through the board) — MIRROR when generating toolpaths. White = milled gap (≥0.4 mm). Copper islands = nets. Component bodies ghosted blue.

This is a placement-true plan, not a Gerber: verify pad pitches against your actual parts before cutting. Drill 1.0 mm (1.3 mm at wire pads).



Milling notes: ① 12 islands total — after milling, buzz out EVERY neighbouring pair for isolation before stuffing ② the OUT and RAIL islands carry 1.4 A: leave them as drawn (wide pours), never “optimize” them into traces ③ fly pads (large) take the FET leads and off-board wiring — drill 1.3 mm ④ power resistors (R_s1/R_s2/R_z) mount 2 mm proud on formed leads ⑤ trimmer RV: pin 1 → VBIAS2 island, pins 2+3 → RVBOT island (check your 3296 pin style!) ⑥ Cb_filt polarity: (+) on VBIAS2, (−) on OUT — it sits across two islands ⑦ islands and pitches match the rev A values incl. Rb_bot select — a different Rb_bot value fits the same pads ⑧ TP1–TP4: solder a wire loop into each pad

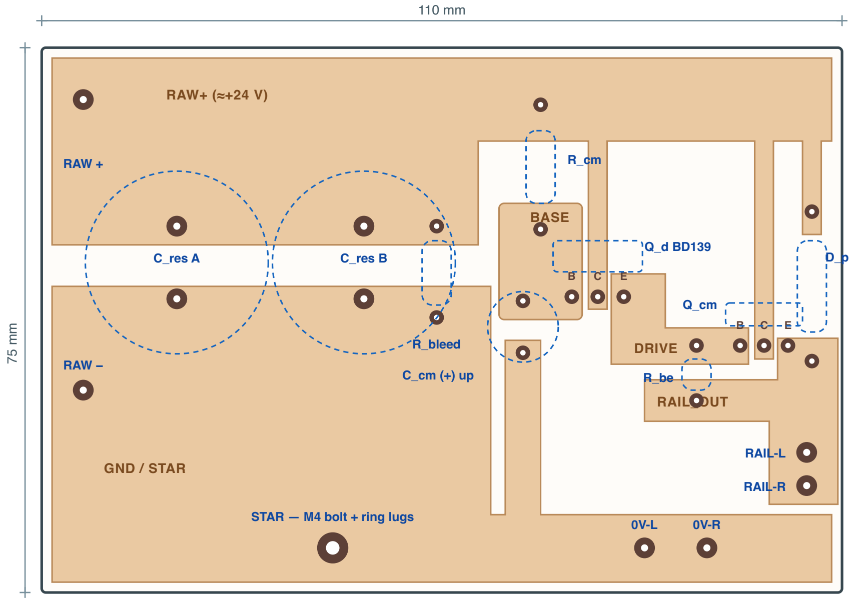
Nets, clockwise from top-left: RAIL · G2 · S2 · OUT(+arm) · S1 · PGND/GND · ZX · IN · G1 · G1GATE · VBIAS2 · RVBOT — cross-check each against Fig. 2 before power.

SONUS MINIMAL	Fig. 9 — Channel board milling plan (single-sided) · rev A	2026-07-03 · LABBOOK App. E

Fig. 10 — PSU Board, isolation-milling plan (single-sided FR4)

Viewed from the COMPONENT side (x-ray) — MIRROR for toolpaths. 5 islands: RAW+ · GND · BASE · DRIVE · RAIL_OUT. Bridge BR1 stays off-board (chassis).

Verify pad pitches against your actual parts before cutting. Drill 1.0 mm; 1.6 mm at snap-in and wire pads; 4.5 mm at the star bolt.



Milling notes: ① only 5 islands — buzz out all adjacent pairs after milling ② RAW+ and GND carry 2.8 A: wide pours stay wide. Of the three narrow RAW+ fingers, two carry only base/leakage current; the middle one (Q_cm collector) carries the full 2.8 A — reinforce it with tinned bus wire laid on the copper ③ Q_cm's heatsink bracket bolts right at the board edge; insulate the tab (TO-220 kit) ④ snap-in caps: 1.6 mm drills, push fully home ⑤ D_p band faces UP (to RAW+ finger) ⑥ star bolt lug stack, board up: C_res returns → RAW– → channel 0 V → speaker returns → chassis bond LAST ⑦ B-C-E letters mark the transistor pads — match them to the device, not to habit

Cross-check every island against Fig. 3 with a continuity beeper BEFORE fitting the transistors.

SONUS MINIMAL	Fig. 10 — PSU board milling plan (single-sided) · rev A	2026-07-03 · LABBOOK App. E

Appendix F — Full references

1. N. Pass, *The Amp Camp Amp*, AudioXpress (2012); build guide at diyaudio.com (ACA) and articles at firstwatt.com.
2. N. Pass, *Zen Amp* (The Audio Amateur, 1994) and *Zen Variations 1–9*, passdiy.com.
3. P. Horowitz, W. Hill, *The Art of Electronics*, 3rd ed., Cambridge UP, 2015 — §§2.2.5, 2.4, 3.1–3.2, ch. 9.
4. D. Self, *Audio Power Amplifier Design*, 6th ed., Focal Press, 2013 — Class-A and grounding chapters.
5. R. Elliott, *Project 15 — Capacitance Multiplier*; *Heatsink Design & Transistor Mounting*; *Death of Zen*; sound-au.com.
6. Infineon/IR IRFP150N datasheet; onsemi MJE15032 datasheet; BD139 datasheet (ST/onsemi); Amphenol Thermometrics CL-60 datasheet; Hammond 1182G18 datasheet (wiring table).
7. REW — Room EQ Wizard (roomeqwizard.com); ARTA (artalabs.hr) — soundcard measurement.
8. Rubycon USC / Nichicon UPW series datasheets (ripple-current ratings used in ECR-3/6).

LABBOOK rev A — generated from the Sonus Minimal handoff (rev 3 BOM) with ECR-1…8 applied. Figures: `diagrams/fig01…fig10`*. Authoritative artifacts listed in §4. Errata: correct this file, bump the rev letter, note the change here.*